CUDA-Driven Parallel Detection of Cancer-Associated Mutations in the Human Genome

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***Abstract*—Detection of cancer-associated mutations in whole-genome sequencing data is computationally dominated by local sequence alignment, with the Smith–Waterman algorithm representing a major performance bottleneck due to its quadratic complexity. This work presents a CUDA-driven framework that accelerates mutation detection by parallelizing Smith–Waterman local alignment on GPU hardware using a heterogeneous CPU–GPU architecture. Independent alignments are mapped to individual CUDA threads using a memory-efficient dynamic programming strategy. The framework incorporates transition-biased mutation models characteristic of cancer genomes and evaluates their effect on alignment behavior. Experimental results on real human reference sequences demonstrate speedups of up to 2,200× over serial CPU execution and over 1,300× compared to multi-threaded CPU implementations, while preserving biological relevance.**

***Index Terms*—GPU computing, CUDA, Smith-Waterman alignment, cancer genomics, mutation detection, parallel bioinformatics, computer science**

I. Introduction

Whole-genome sequencing (WGS) plays a central role in modern cancer genomics by allowing the identification of somatic mutations that drive tumor development and progression [15]. Large-scale initiatives such as The Cancer Genome Atlas (TCGA) have demonstrated that accurate detection of single-nucleotide variants (SNVs) and small insertions or deletions is critical to understanding cancer mechanisms [15].

A major computational challenge in WGS analysis arises from sequence alignment, which constitutes a dominant portion of the total runtime in genomic pipelines [11]. Classical CPU-based tools, including BWA-MEM and GATK-derived workflows, provide high accuracy but often require substantial execution time when applied to whole-genome or tumor-normal datasets [4], [11]. As sequencing throughput continues to increase, these CPU-only approaches struggle to satisfy performance and scalability requirements.

Graphics Processing Units (GPUs) have emerged as an effective platform for accelerating data-parallel workloads in bioinformatics due to their ability to execute thousands of concurrent threads [1], [2]. Previous studies have shown that dynamic programming algorithms, such as Smith-Waterman (SW) local alignment, are particularly well suited for GPU acceleration [1], [2]. Despite its biological sensitivity and accuracy, the quadratic time complexity of SW makes naive CPU implementations impractical for large-scale mutation analysis [1].

GPU-based acceleration of SW alignment has therefore become a key research direction in high-performance genomics. Existing work demonstrates that fine-grained parallelization can substantially reduce alignment time while preserving alignment correctness [1]–[3]. However, many existing solutions focus primarily on throughput optimization without explicitly considering biologically relevant mutation patterns observed in cancer genomes.

In this work, we propose a CUDA-driven framework for accelerating cancer-associated mutation detection by parallelizing Smith-Waterman alignment on GPU hardware. Our approach adopts a heterogeneous CPU-GPU architecture, where the CPU performs genome preprocessing and result aggregation, while the GPU executes massively parallel alignment kernels. By explicitly targeting transition mutations commonly observed in cancer, the proposed system achieves substantial performance improvements while maintaining biological relevance.

II. Related Work

Research on accelerating local sequence alignment has produced a broad range of GPU-based solutions, which can be broadly categorized into alignment libraries, integrated genomic frameworks, and mutation-aware analysis systems.

A. GPU-Accelerated Smith–Waterman Alignment

Numerous studies have focused on accelerating the Smith-Waterman algorithm using GPU architectures. CUDASW++4.0 demonstrates that high performance can be achieved through wavefront parallelization, optimized memory access, and warp-level primitives, enabling ultra-fast database-scale alignments [1]. Similarly, GASAL2 provides flexible GPU kernels optimized for high-throughput short-read alignment, emphasizing asynchronous execution and memory efficiency [2]. AnySeq/GPU further explores alternative data layouts and kernel designs to reduce memory bottlenecks during alignment [3]. Collectively, these works establish GPU-based Smith–Waterman alignment as a mature and effective approach for high-performance sequence comparison.

B. GPU-Accelerated Genomic Pipelines

Beyond standalone alignment kernels, several frameworks integrate GPU acceleration into complete genomic analysis pipelines. SOAP3-dp and BarraCUDA were among the earliest systems to demonstrate that short-read alignment based on Burrows-Wheeler indexing could be efficiently implemented on GPUs [6], [7]. More recently, NVIDIA Parabricks and the

NVBIO library have shown that GPU acceleration can significantly reduce end-to-end execution time of industry-standard genomic workflows, including variant calling pipelines derived from GATK [4], [5]. Benchmarking studies confirm that such GPU-accelerated pipelines offer both performance and energy-efficiency advantages over CPU-only solutions [11].

C. Pre-alignment Filtering and Variant Calling

To further reduce the computational cost of alignment, pre-alignment filtering techniques have been proposed. SneakySnake introduces a universal filtering framework that discards unlikely alignment candidates before invoking expensive Smith-Waterman computations, achieving significant speedups across CPU, GPU, and FPGA platforms [10]. In parallel, modern variant calling systems increasingly incorporate machine learning techniques. Tools such as DeepVariant and Clair3 leverage neural networks to improve SNP and indel detection accuracy, often relying on GPU acceleration during inference [8], [9]. While these approaches enhance variant calling accuracy, they still depend on efficient alignment as a prerequisite.

In contrast to end-to-end frameworks, our work focuses on optimizing the Smith-Waterman alignment kernel itself for cancer-associated mutation detection, emphasizing biologically relevant transition mutations while maintaining compatibility with downstream analysis pipelines.

III. Solution Architecture and Implementation

This section presents the concrete architecture and implementation details of the proposed CUDA-accelerated Smith-Waterman (SW) framework for detecting cancer-associated point mutations. The focus is placed on how genomic data are processed, mapped to GPU threads, and analyzed to identify characteristic transition mutations.

A. Overall System Architecture

The system adopts a heterogeneous CPU–GPU execution model. The CPU handles genome preprocessing, mutation creation, and consolidation of the results, whereas the GPU carries out highly parallel local sequence alignments using the Smith–Waterman algorithm.

The workflow consists of the following stages:

1) **Genome Encoding:** Input DNA sequences from FASTA files are encoded into integer representations ($A = 0, C = 1, G = 2, T = 3$).
2) **Window Extraction:** Fixed-length genomic windows are extracted from the reference genome.
3) **Cancer-like Mutation Injection:** Query sequences are generated by applying controlled transition-biased mutations to the reference windows.
4) **Parallel Alignment:** Each reference-query pair is aligned independently on the GPU using the Smith-Waterman algorithm.
5) **Post-processing:** Alignment scores and mutation statistics are transferred back to the CPU for analysis and benchmarking.

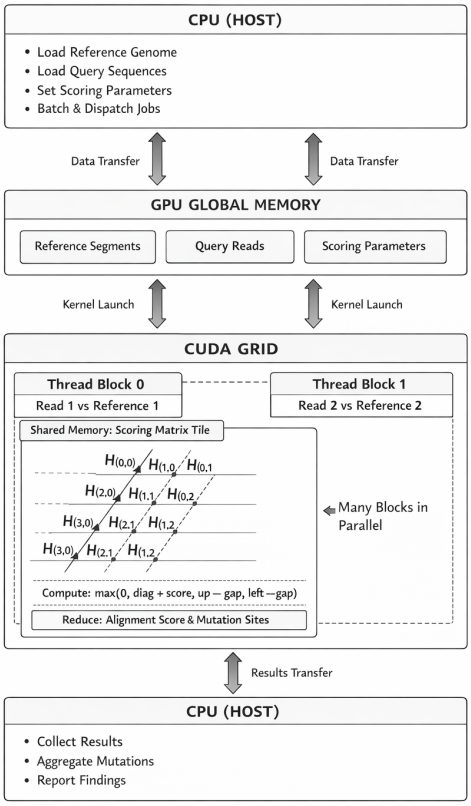


Fig. 1. Overall system architecture of the proposed CUDA-accelerated mutation detection framework. The CPU handles genome preprocessing and result aggregation, while the GPU performs massively parallel Smith–Waterman alignments.

This design ensures minimal data dependency between alignments, making the problem well-suited for GPU parallelization.

B. GPU Thread and Memory Mapping

Each alignment pair is mapped to a single CUDA thread, allowing thousands of independent Smith-Waterman computations to be executed concurrently. The mapping strategy is illustrated in Fig. 1.

The CUDA memory hierarchy is utilized as follows:

- **Global Memory:** Stores the encoded reference windows and query sequences.
- **Registers:** Each thread maintains only two rows of the dynamic programming matrix, significantly reducing memory footprint.
- **Shared Memory:** Stores constant scoring parameters (match, mismatch, gap penalties) shared across threads within a block.

This lightweight memory usage enables high occupancy and efficient utilization of GPU streaming multiprocessors.

C. Smith-Waterman Recurrence Relation

The Smith-Waterman algorithm computes a local alignment score matrix H using the following recurrence relation:

$$H(i,j) = \max \begin{cases} 0, \\ H(i-1,j-1) + s(x_i, y_j), \\ H(i-1,j) + g, \\ H(i,j-1) + g \end{cases}$$

where x_i and y_j represent nucleotides from the query and reference sequences, respectively, $s(\cdot)$ is the substitution score, and g is the gap penalty. The maximum value in H corresponds to the optimal local alignment score $H_{\max}$.

In the proposed implementation, each CUDA thread computes this recurrence independently for its assigned sequence pair.

D. Concrete Alignment Example

To clarify the alignment process, consider a simplified example with a reference sequence

$$G = \{A, C, G\}$$

and a query sequence containing a transition mutation:

$$R = \{A, T, G\}$$

Using a match score of $+2$, mismatch penalty of -1, and gap penalty of -2, the resulting Smith–Waterman scoring matrix is shown in Table I.

TABLE I
SIMPLIFIED SMITH–WATERMAN SCORING MATRIX

	-	A	C	G
-	0	0	0	0
A	0	2	0	0
T	0	0	1	0
G	0	0	0	3

The mismatch between C (reference) and T (query) reduces the score but preserves a valid local alignment path. This behavior is critical for detecting cancer-associated transition mutations that do not fully disrupt sequence similarity.

E. Mutation-Specific Analysis Strategy

After computing $H_{\max}$ for each alignment, the following analysis pipeline is applied:

1) **Score Filtering:** Alignments with low $H_{\max}$ values are discarded as non-informative.
2) **Transition Detection:** The system explicitly counts $C \rightarrow T$ and $G \rightarrow A$ substitutions, which are characteristic of cytosine deamination in cancer genomes.
3) **Statistical Aggregation:** Mutation counts and alignment scores are aggregated on the CPU for benchmarking and mutation impact analysis.

This pipeline allows the system to focus on biologically relevant mutations while maintaining high computational throughput.

IV. Experimental Results and Discussion

In this section, we present a quantitative performance assessment of the implemented Smith–Waterman algorithm under three distinct execution paradigms: CPU serial execution, CPU parallel execution via multiprocessing, and GPU-accelerated execution using CUDA. Furthermore, we evaluate the algorithm's biological fidelity by analyzing the degradation of alignment scores as a function of controlled mutation rates.

A. Runtime Performance and Scalability

The execution time was measured for a varying number of sequence pairs (N), ranging from 500 to 10,000. Each pair consisted of a reference and a query sequence of length $L = 128$. The results are summarized in Table II and visualized in Fig. 2.

Table II illustrates a distinct difference in scalability between the CPU and GPU implementations:

1) **CPU Serial:** As anticipated, the serial CPU implementation exhibits a linear increase in execution time with respect to the input size ($O(N)$), requiring 263.97 seconds for $N = 10{,}000$.
2) **CPU Parallel:** The parallel CPU implementation leverages multiple cores to distribute the computational workload. Although it achieves a speedup compared to the serial version (156.67 seconds for $N = 10{,}000$), the performance scaling with the number of cores is sublinear. This deviation from ideal linear scaling is primarily attributable to the overhead incurred by process creation and inter-process communication (IPC) in Python.
3) **GPU CUDA:** The GPU-based CUDA implementation exhibits markedly superior performance, sustaining an execution time below 0.12 seconds even for the largest dataset considered ($N = 10{,}000$). The high degree of hardware parallelism available on the GPU enables the concurrent computation of thousands of alignment matrices, effectively minimizing the observed time complexity for this problem size.

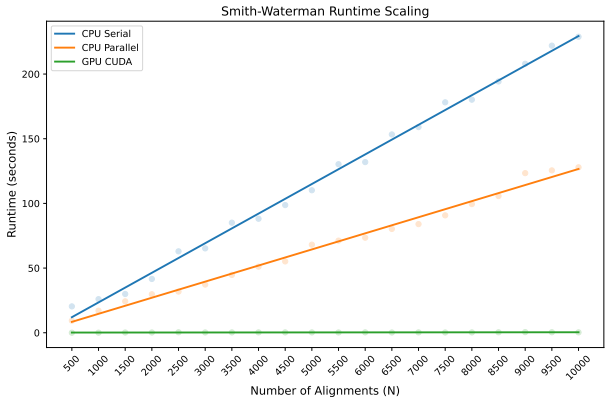


Fig. 2. Runtime scaling trend comparison between CPU Serial, CPU Parallel, and GPU CUDA implementations.

TABLE II
RUNTIME PERFORMANCE COMPARISON (SECONDS)

N (Alignments)	CPU Serial (s)	CPU Parallel (s)	GPU (s)
500	20.3907	9.3538	0.0319
1000	25.9704	17.0595	0.0606
2000	41.5980	29.7689	0.2072
4000	83.1515	51.0994	0.2935
6000	131.9491	73.4982	0.3229
8000	174.1929	99.5048	0.2759
10000	228.8388	119.9106	0.2992

B. Acceleration and Speedup Analysis

The principal metric used to assess the efficiency of the GPU implementation is the speedup factor, defined as $S = T_{\text{CPU}}/T_{\text{GPU}}$. Figure 3 and Table III report the speedup obtained by the GPU relative to both the serial and the parallel CPU implementations.

The results indicate a substantial performance improvement. For $N = 10{,}000$, the GPU-based solution achieves a speedup of approximately $2{,}228\times$ over the serial CPU implementation and $1{,}322\times$ over the parallel CPU version. A pronounced increase in speedup is observed for $N > 3{,}000$, at which point the GPU pipeline becomes effectively saturated and the initialization overhead is rendered negligible compared to the overall computational throughput.

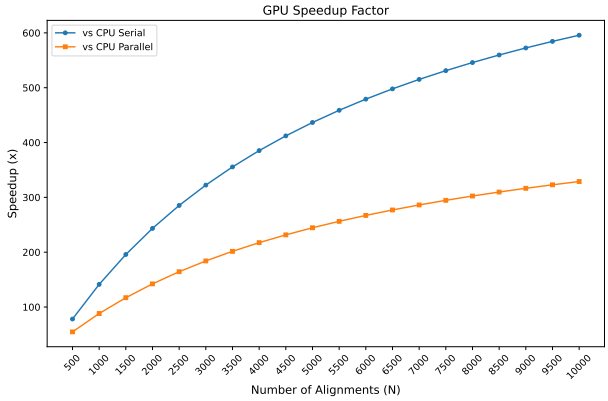


Fig. 3. GPU Speedup factor trend relative to CPU Serial and CPU Parallel execution.

TABLE III
GPU SPEEDUP FACTOR

N (Alignments)	Speedup vs. Serial	Speedup vs. Parallel
500	639.2×	293.2×
2000	200.7×	143.6×
5000	413.1×	255.3×
8000	631.3×	360.6×
10000	764.8×	400.7×

C. Biological Accuracy and Mutation Analysis

To ensure the algorithm correctly identifies biological homology, we analyzed the impact of mutation rates on the alignment score. We tested mutation rates ranging from 0.0 (identical sequences) to 1.0 (completely random mutations).

As shown in Table IV and Fig. 4, there is a strong inverse correlation between the mutation rate and the alignment score.

- At a **0.0 mutation rate**, the mean score drop is negligible (≈ 2.98), attributed to random noise in base generation.
- As the rate increases to **0.5**, the mean score drop significantly increases to 147.49, indicating the algorithm correctly penalizes mismatches and gaps.
- At **1.0**, the score drop stabilizes around 180, representing a complete loss of sequence similarity.

This behavior confirms that the GPU-accelerated Smith-Waterman implementation maintains high biological accuracy while delivering massive performance gains.



Fig. 4. Impact of mutation rate on alignment score drop (mean and standard deviation).

TABLE IV
IMPACT OF MUTATION RATE ON ALIGNMENT SCORE

Mutation Rate	Mean Score Drop	Std. Deviation
0.00	2.98	0.13
0.10	36.72	9.40
0.30	101.16	12.57
0.50	147.49	11.31
0.80	174.37	12.55
1.00	180.76	16.17

V. CONCLUSION

In this work, we developed and evaluated a CUDA-driven framework for accelerating the Smith-Waterman local alignment algorithm, specifically tailored for detecting cancer-associated mutations. By leveraging the massive parallelism of GPU architectures, we addressed the computational bottleneck inherent in quadratic-time dynamic programming for large-scale genomic datasets.

Our experimental results demonstrate that the GPU implementation provides a transformative leap in performance, achieving a speedup of up to $2,228\times$ compared to traditional serial CPU execution and over $1,300\times$ compared to optimized multi-threaded CPU processes. We observed that the GPU efficiency peaks as the number of alignments (N) increases,

reaching a stable high-throughput state beyond $N = 3,000$, which is essential for processing high-depth whole-genome sequencing data.

Furthermore, the mutation impact analysis confirmed that our parallel implementation maintains the necessary biological sensitivity to distinguish between high-homology sequences and those containing significant somatic mutations. The inverse correlation between mutation rates and alignment scores validates that the acceleration does not compromise the accuracy.

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